STAFF REVIEW · H446 067 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Breaking a complex problem into smaller, manageable parts. The parts may depend on one another.

2. Sorted elements.

3. Returns the smaller argument.

4. low0 high6 mid3=8, then low4 high6 mid5=12; found index5.

I Do — source worked steps

1. Follow the reasoning — Each iteration narrows the candidate interval, and result is index 6.

We Do — guided CHECK

1. Follow the reasoning
The second comparison finds the target after removing upper half.

You Do — source mark schemes

1. mid3=10 [2], low4 then mid5=16 [2], low6 mid6=19 [2].

2. Split [7,2] and [9,1] [2], solve [2,7] and [1,9] [2], merge [1,2,7,9] [2].

3. Linear O(n), up to1024 checks [2]; binary O(log n), at most ~11 candidate checks with conventional bounds [2]; sorted data required [2].

Exit Ticket — source answer

1. False. Sorting is required.